Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/05/2025
PI Name:	Boppart, Marni		
Project Title:	Design of an Extracellular Vesicle Approach to Protect Human Health in Space		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	Urbana	**State:**	IL
Zip Code:	61801	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 TRISH Biomedical Research Advances for Space Health (BRASH)
Start Date:	07/01/2023	**End Date:**	06/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	6
No. of PhD Candidates:	9	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Burd, Nicholas (University Of Illinois, Urbana-Champaign) Kong, Hyunjoon (University Of Illinois, Urbana-Champaign) Rhodes, Justin (Self) Saif, M Taher (University Of Illinois, Urbana-Champaign) Sweedler, Jonathan (University Of Illinois, Urbana-Champaign)		
Grant/Contract No.:	NNX16AO69A-T0701		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Exercise has been embraced by astronauts to reduce tissue degeneration; yet diminished mechanical cues prevent full preservation of tissue mass and function during long-term spaceflight. Therefore, autologous supplementation with the exercise secretome may represent an ideal disruptive approach to improve endogenous repair. Based on our preliminary results, demonstrating the capacity for ExerVs to promote consistent and visible increases in vascular growth in murine skeletal muscle, we propose that ExerVs may be collected under normal gravitation force, cryopreserved, and reintroduced prior to and during spaceflight. We will accomplish our goal by pursuing the following two Aims: (1) to demonstrate the capacity for human-derived ExerVs to improve endogenous repair and growth mechanisms in vitro and in vivo, and (2) to demonstrate the capacity for artificially manufactured ExerVs to improve endogenous repair and growth mechanisms in vitro and in vivo. As directed by the solicitation, we will evaluate the impact of natural and artificial human-derived ExerVs on multiple organ systems, including vessels, muscle and brain, in collaboration with colleagues in Neuroscience, Mechanical Science and Engineering, and Chemical and Biomolecular Engineering. Using our innovative and integrative approach, we are confident we will deliver a novel product that will effectively preserve human health in space in a controlled and safe manner.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The primary objective of this study is to collect extracellular vesicles (EVs) from blood of human subjects after exercise (ExerVs) and test their ability to preserve tissue mass and function during disuse in mice (Aim 1). If successful, artificial nanoparticles will be manufactured based on ExerV content for further evaluation in mice and future human clinical trials (Aim 2). The results from this study indicate that ExerVs are enriched in metabolite interconversion enzymes, specifically oxidoreductases, including glutathione peroxidase 1 (GPX1). ExerVs collected from male subjects were not able to preserve skeletal muscle mass during disuse in a mouse model of microgravity (hindlimb suspension), but were able to prevent collagen accumulation in a manner that allowed for robust regrowth of Type 1 oxidative muscle fibers and capillary content during the period of reloading. GPX1-encapsulated liposomes were able to recapitulate these effects. ExerVs did not significantly impact brain volume or neurogenesis during or after disuse. Finally, proteomic analysis suggests that female ExerVs are enriched in diverse molecular cargo compared to male ExerVs, and thus possess higher potential for preservation or recovery of tissue mass. Overall, this study suggests that ExerVs and/or manufactured liposomes may provide a novel solution to preserving whole body plasticity and tissue mass during spaceflight.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Connolly MG, Fliflet AM, Ravi P, Rosu DI, Boppart MD, Rhodes JS. "Exercise-induced plasma-derived extracellular vesicles increase adult hippocampal neurogenesis." Brain Research. 2025 Dec 15;1869:150003. https://doi.org/10.1016/j.brainres.2025.150003 , Dec-2025
Articles in Peer-reviewed Journals	Fliflet AM, Tan Y, Barnes TM, Vijayan AN, Choi SJ, Deutz MT, Zupancic Z, Miller BF, Burd NA, Sweedler JV, Boppart MD. "Human plasma extracellular vesicles as an exercise mimetic to preserve skeletal muscle plasticity during disuse." npj Microgravity. 2026 Mar 6. Early access article.https://doi.org/10.1038/s41526-026-00582-4 ; PMID: 41792160 , Mar-2026
Articles in Peer-reviewed Journals	Fliflet AM, Spradlin RA, Tan Y, Nishitha Vijayan A, Choi SJ, Kao WC, Aksamitiene E, Nelappana M, Ding S, Huang KY, Joshi A, Kambar N, Bludgen T, Meehan M, Boss JL, Knipp M, Fan H, Leal C, Sunde RA, Dobrucki LW, Boppart SA, Kong HJ, Sweedler JV, Boppart MD. "Exercise training stimulates the release of glutathione peroxidase 1 (GPX1)-enriched extracellular vesicles that promote angiogenesis." Faseb j. 2026 Jun 30;40(12):e72052. https://doi.org/10.1096/fj.202505096RR ; PMID: 42313915; PMCID: PMC13278521 , Jun-2026
Awards	Sweedler J. "Beckman Institute Vision and Spirit Award." Apr-2025
Awards	Sweedler J. "Elected to the American Academy of Arts & Sciences." May-2025
Awards	Boppart M. "College of Applied Health Sciences, Dean Fellow." Aug-2024
Awards	Boppart M. "Faculty Ranked as Excellent for Teaching, UIUC." May-2024
Awards	Boppart M. "Presidential Early Career Award for Scientists and Engineers (PECASE)." Jan-2025
Awards	Saif T. "Elected to National Academy of Engineering." May-2025
Awards	Saif T. "Grainger Distinguished Chair." May-2025